Tool Survey & Requirements Specification Report

Linux Network Monitor & Controller

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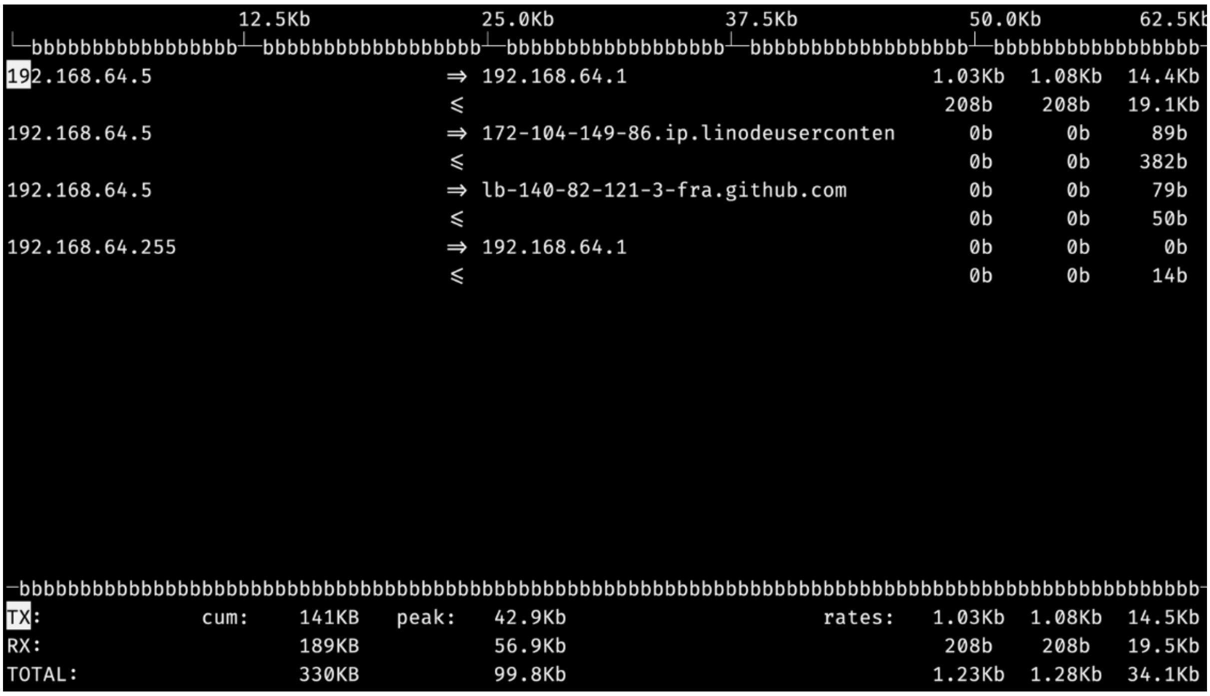
1. Tool Survey & Comparative Analysis

1.1 Iftop



iftop is a passive monitoring tool. This program monitors network traffic and displays a table of current bandwidth usage, bandwidth measured in bits per second at default, and may be changed to bytes per seconds with the command line option.

Strengths

- Real-time traffic visibility

- Lightweight and easy to use
- Supports filtering and sorting
- Clear connection-based view

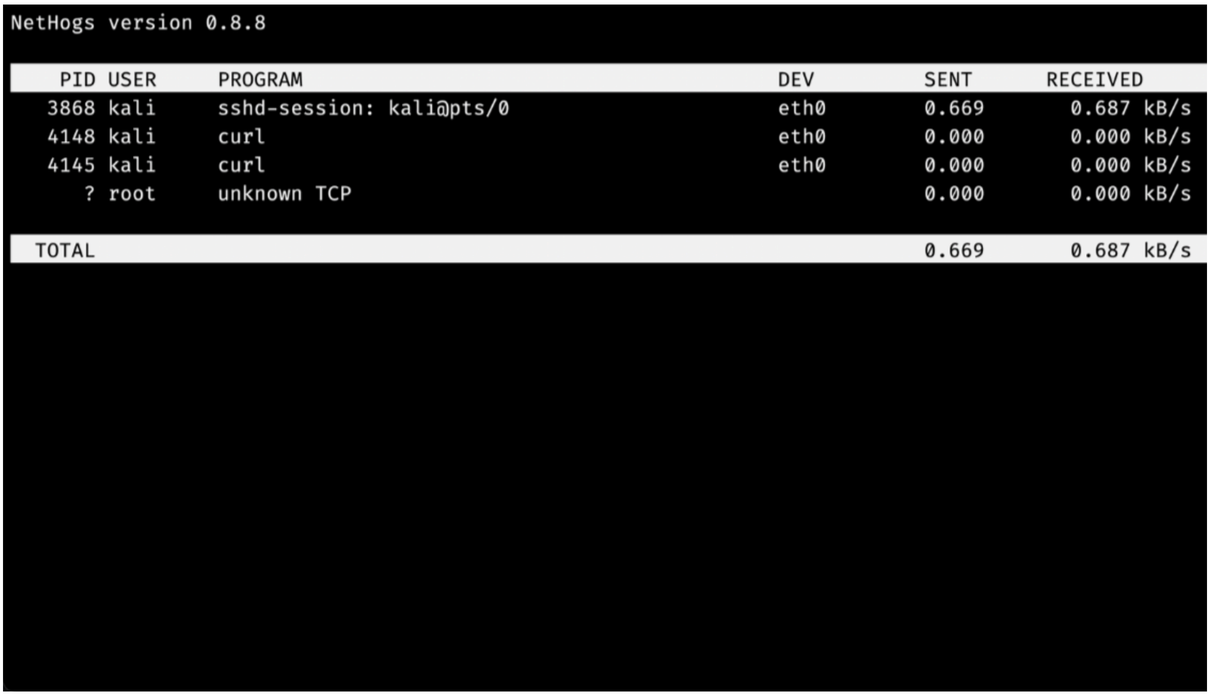
Limitations

- No process-level monitoring
- No historical data
- Single interface focus
- No GUI

Takeaway

Iftop is excellent for answering *“who is talking to whom right now”*, but it is limited for deeper system analysis.

1.2 NetHogs



NetHogs is a small 'net top' tool. Instead of breaking the traffic down per protocol or per subnet, like most tools do, it groups bandwidth by process.

Strengths

- Per-process bandwidth tracking
- Real-time monitoring
- Simple and practical

Limitations

- No historical data
- Limited filtering and features
- No GUI
- Not suitable for large-scale monitoring

Takeaway

NetHogs introduces the key idea of mapping network usage to processes, which is essential for system-level understanding.

1.3 VnStat

```
┌──(kali㉿kali)-[~]
└─$ vnstat
Database updated: 2026-03-28 09:50:00

   eth0 since 2026-03-27

          rx:  5.39 MiB      tx:  2.42 MiB      total:  7.82 MiB

   monthly
                     rx      |     tx      |    total    |   avg. rate
     ------------------------+-------------+-------------+---------------
       2026-03      5.39 MiB |    2.42 MiB |    7.82 MiB |    1.77 kbit/s
     ------------------------+-------------+-------------+---------------
     estimated     50.85 MiB |   22.75 MiB |   73.60 MiB |

   daily
                     rx      |     tx      |    total    |   avg. rate
     ------------------------+-------------+-------------+---------------
     yesterday      3.84 MiB |    1.57 MiB |    5.41 MiB |      524 bit/s
         today      1.56 MiB |  875.18 KiB |    2.41 MiB |      571 bit/s
     ------------------------+-------------+-------------+---------------
     estimated      3.80 MiB |    2.08 MiB |    5.88 MiB |
```

vnStat is an open source network utility for Linux and BSD. It uses a command line interface. vnStat command is a console-based network traffic monitor. It keeps a log of hourly, daily and monthly network traffic for the selected interface(s) but is not a packet sniffer. The traffic information is analyzed from interfaces statistics provided by the kernel. That way vnStat can be used even without root permissions.

Strengths

- Historical data storage
- Very low resource usage
- Supports multiple interfaces
- Useful for long-term analysis

Limitations

- No real-time deep inspection
- No process-level visibility
- Requires setup (database/daemon)
- No GUI

Takeaway

VnStat is ideal for tracking trends over time but not for real-time debugging.

1.4 Overall Comparison

- **Iftop (connections Analyzer) (real-time)**
- **NetHogs (processes Inspection) (real-time)**
- **VnStat (historical data)**

Key Insight:

No single tool combines all three aspects. This gap motivates building a system that integrates:

- real-time monitoring
- process correlation

- historical analysis

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2. Requirements Specification

2.1 Functional Requirements

The system shall:

- Monitor network traffic in real time
- Display active connections (IP, port, protocol, bandwidth)
- Map traffic to processes (PID) and users
- Provide a GUI dashboard for visualization
- Support filtering and sorting (process, user, protocol, usage)
- Show top bandwidth-consuming processes
- Generate alerts for abnormal activity
- Store traffic data for later analysis

These combine features inspired by Iftop, NetHogs, and VnStat into one system.

2.2 Non-Functional Requirements

Performance

- **Display update within 500 ms**
  - 500 ms was chosen as a balance between real-time responsiveness and keeping CPU and GUI overhead reasonable and based on our research on the famous tool WireShark.
- **Handle packet capture with less than 1% loss at moderate traffic levels**
  - This target keeps the monitor accurate under normal use while staying realistic for a user-space course project.
- **CPU usage below 15% during typical monitoring**
  - Since the tool may run continuously, it should stay lightweight and not become a noticeable burden on the monitored system.

Scalability

- **Monitor 100–500 simultaneous connections efficiently**
    - This range was selected to represent moderate real-world usage while still being achievable within the project scope.

Reliability

- Handle malformed packets and traffic spikes
- Stable under continuous monitoring

Security

- Minimal privilege escalation
- Read-only packet handling

Usability

- Intuitive GUI and CLI
- Easy filtering and navigation

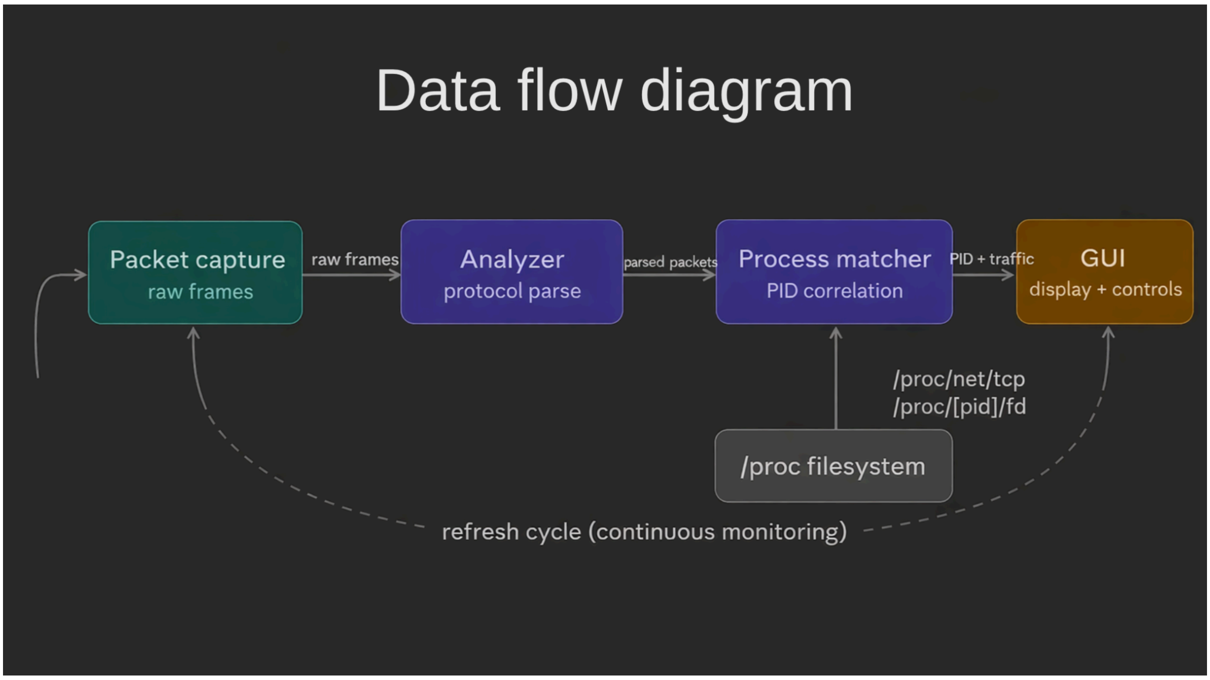
2.3 System Constraints

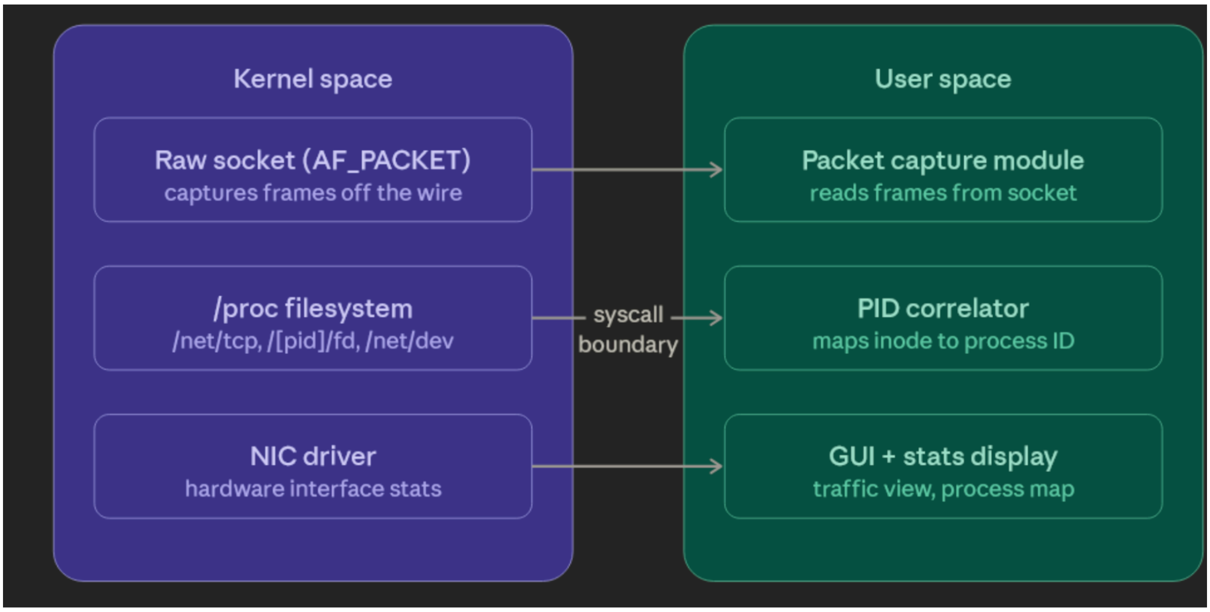
- Linux-only system
- Requires root access for packet capture
- Real-time processing requires frequent refresh (~500 ms)
- Must optimize CPU usage (caching, incremental updates)
- GUI must remain responsive under heavy data

2.4 Architectural Assumptions

- Modular design (independent components)

- Pipeline data flow:
    - Packet capture → parsing → process mapping → GUI
- Linux kernel provides reliable process-port mapping
- Accurate system time (NTP) for logging





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3. Conclusion

The analysis of existing tools shows that each solves part of the problem:

- Iftop → real-time connections
- NetHogs → process-level insight
- VnStat → historical tracking

The proposed system aims to combine all these capabilities into a single, efficient, and user-friendly solution with a modern GUI and strong system-level integration.